



Asthma and Air Pollution: Protecting Children's Health

Abstract

Nearly 1 in 10 American children has asthma. Air pollution from vehicles, power plants, and wildfires can increase a child's risk of developing asthma and can trigger attacks, and climate change is making several of these triggers worse. This lesson, based on a Harvard Chan C-CHANGE guide, explores the evidence and why children are especially vulnerable.

Introduction

Asthma is a chronic lung disease that can make it hard to breathe. People with asthma may cough, wheeze, feel chest tightness, or struggle to catch their breath. Burning fossil fuels in power plants, cars, and trucks produces carbon pollution that drives climate change, and it also creates conventional air pollution, including fine particulate matter (PM), that harms health. Long-term exposure to air pollution can significantly increase the risk of a child developing asthma, especially in young children, and breathing polluted air can trigger attacks in people who already have asthma. As the climate warms, pollution, pollen, and wildfire smoke are all getting worse.

How Do Scientists Study This?

Researchers use several kinds of studies to link air pollution and climate to asthma. Cohort studies follow groups of children over time: one study of 1,200 children in Boston tracked their lifetime exposure to black carbon (soot) and particulate matter and compared it with whether they developed asthma by ages 3 to 5. Health records studies analyze large numbers of hospital visits: one study examined 170,000 emergency and urgent care visits in San Diego from 2011 to 2017 to compare wildfire smoke with other pollution sources. Government assessments, such as a 2023 EPA report, combine this evidence with climate projections to estimate future asthma cases.

How climate change worsens asthma



Follow children

Cohort studies track pollution exposure and asthma in the same children over years.

Analyze visits

Researchers compare emergency visits on smoky and polluted days with other days.

Project the future

Assessments combine health evidence with climate projections to estimate new cases.

What Did They Discover?

In the Boston study, lifetime exposure to soot and particulate matter was linked to asthma in early childhood. In San Diego, wildfire smoke was far more toxic to children's lungs than other air pollution: increases in emergency visits were 10 times higher for wildfire smoke than for other sources. The 2023 EPA report projected that climate-driven warming of 2°C to 4°C would increase annual childhood asthma cases by 4% to 11%. Climate change worsens asthma in several ways: higher temperatures create more ground-level ozone, a powerful lung irritant; longer pollen seasons trigger allergic asthma; and hotter, drier conditions fuel more frequent and intense wildfires whose smoke can travel across the country.

Asthma and air pollution

~1 in 10

American children have asthma

10×

higher ER visit increases from wildfire smoke

4–11%

more childhood asthma cases at 2–4°C of warming

Children are more vulnerable than adults because their lungs are still developing and they breathe faster, taking in more pollution for their size.

Takeaways

Pollution causes asthma

Long-term exposure to soot and particles raises children's asthma risk.

Smoke is especially toxic

Wildfire smoke drove 10 times more emergency visits than other pollution.

Climate adds triggers

Warming increases ozone, pollen, and wildfire smoke that worsen asthma.

Key Terms

Asthma

A chronic lung disease in which airways swell and narrow, making it hard to breathe.

Particulate matter (PM)

Tiny particles in the air, such as soot, that can travel deep into the lungs and bloodstream.

Ground-level ozone

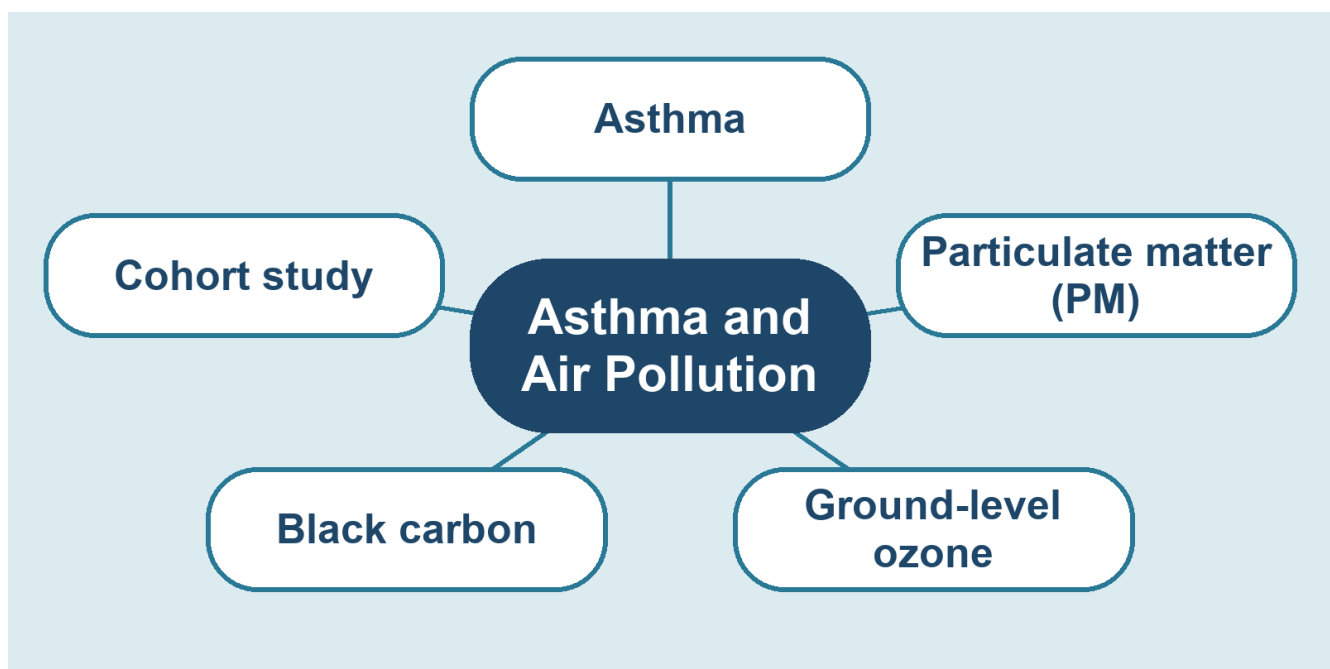
A lung-irritating gas that forms when pollution reacts in sunlight and heat; it increases on hot days.

Black carbon

Soot produced by burning fuels, a common part of particulate air pollution from traffic.

Cohort study

A study that follows a group of people over time to see how exposures relate to health outcomes.



Activity

Asthma Trigger Map

Students map asthma triggers around their school and propose changes to protect students with asthma.

Materials: Map or sketch of the school and surrounding streets; Trigger stickers: traffic, idling buses, pollen trees, smoke, indoor irritants; Air quality website such as airnow.gov (optional)

1. Mark major roads, bus drop-off areas, and green spaces on the map.
2. Place trigger stickers where students might be exposed to asthma triggers, indoors and outdoors.
3. Identify which triggers climate change is likely to make worse and explain why.
4. Propose two changes, such as a no-idling zone or indoor recess on high-ozone days.
5. Write a short letter to your principal recommending one change, using evidence from the lesson.

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Asthma and Air Pollution: Protecting Children's Health. <https://hsph.harvard.edu/research/climate-health-c-change/climate-kids-and-health/asthma/>